

Points to Consider when Modelling

Ferran Torres, MD, PhD

Biostatistics Unit. Medical School. Office M3/323B
Universitat Autònoma de Barcelona (UAB)

EXPLANATORY VS PREDICTIVE

Explanatory Versus Predictive Modeling

Kristin L. Sainani, PhD

INTRODUCTION

When building multivariate statistical models, researchers need to be clear as to whether their goals are explanatory or predictive. Explanatory research aims to identify risk (or protective) factors that are causally related to an outcome. Predictive research aims to find the combination of factors that best predicts a current diagnosis or future event. This distinction affects every aspect of model building and evaluation. Unfortunately, researchers often conflate the two, which leads to errors [1]. This article reviews the differences between explanatory and predictive modeling.

Association (not causation!)

Sainani KL.
Explanatory versus predictive modeling.
PM R. 2014 Sep;6(9):841-4

EXPLANATORY MODELING

The aim of explanatory research is to establish causal relationships. For example, in the article by Yu et al [2], the researchers attempted to identify risk factors that would predispose amputees to falling during the postoperative period. A better understanding of these risk factors could help researchers design interventions to prevent falls. Explanatory model building is primarily concerned with identifying individual risk factors that are associated with the outcome as well as ruling out confounding (extraneous variables that are related to both the risk factor and outcome may create spurious associations between them). Explanatory modelers need to worry about chance findings, unmeasured confounding, and residual confounding.

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PREDICTIVE MODELING

Prediction models aim to accurately estimate the probability that a disease is present (diagnosis) or that a future event will occur (prognosis). For example, Bates et al [3] built a model to predict 1-year mortality of veterans with stroke. Knowing a patient's mortality risk can help patients, physicians, and caregivers to better plan postdischarge care and priorities. In predictive modeling, overall predictive accuracy is paramount and the role of individual variables is less critical. Variables may be included in the final model even if they are not causally related to the outcome. For example, stroke patients discharged to an acute care facility are more likely to die, but this variable is a marker of poor health rather than a cause of death.

Predictive modelers need to consider several aspects of model performance. They also need to worry about overfitting and generalizability. Overfit models are tuned to the idiosyncrasies of a particular sample and thus have high predictive accuracy for the sample but not for new observations. Because of the problem of overfitting, prediction models should always undergo validation.

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Summary conclusions

Researchers with **explanatory** goals get sidetracked when trying to optimize overall model metrics (such as R^2 values or areas under the ROC curve) and neglect issues such as confounding.

Researchers with **predictive** goals get sidetracked worrying about individual β coefficients and p-values, and omit critical steps such as calibration or validation.

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Table 1. Summary of explanatory versus predictive models

Explanatory Models		Predictive Models
Goal	Establish causal relationships	Predict current diagnoses or future outcomes
Threats to validity	Chance findings (type I errors); confounding	Overfitting; lack of generalizability to new populations
Candidate variables	A limited set of prespecified risk factors and confounders	A larger set of potential predictors; some predictors may not be causally related to the outcome
Variable selection	Hypothesis driven; should not use automated selection procedures	Exploratory; may use automated selection procedures, but validation is essential and newer automated procedures that incorporate shrinkage are preferred
Measures of model performance	Size of β coefficients for individual risk factors; level of significance for individual risk factors	Discrimination (eg, ROC analysis); calibration (eg, Hosmer-Lemeshow test); goodness of fit (eg, R^2 , AIC); reclassification (eg, net reclassification index); clinical utility
Validation	New studies are needed to confirm individual causal relationships	Internal validation: split-sample validation; cross validation; bootstrap validation; external validation

ROC = receiver operating characteristic; AIC = Akaike information criterion.

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“Grouping variable”

- Case-Control (CC) study:
 - Designed group is **CC** (the outcome result)
$$\text{CC} = \text{Exposure} + \text{Covar}_1 \dots \text{Covar}_x$$
- Cohort Study:
 - Designed group is the **exposure (Cohort)**
$$\text{Outcome} = \text{Cohort} + \text{Covar}_1 \dots \text{Covar}_x$$
- Both:
 - Adjust by confounders $\text{Covar}_1 \dots \text{Covar}_x$

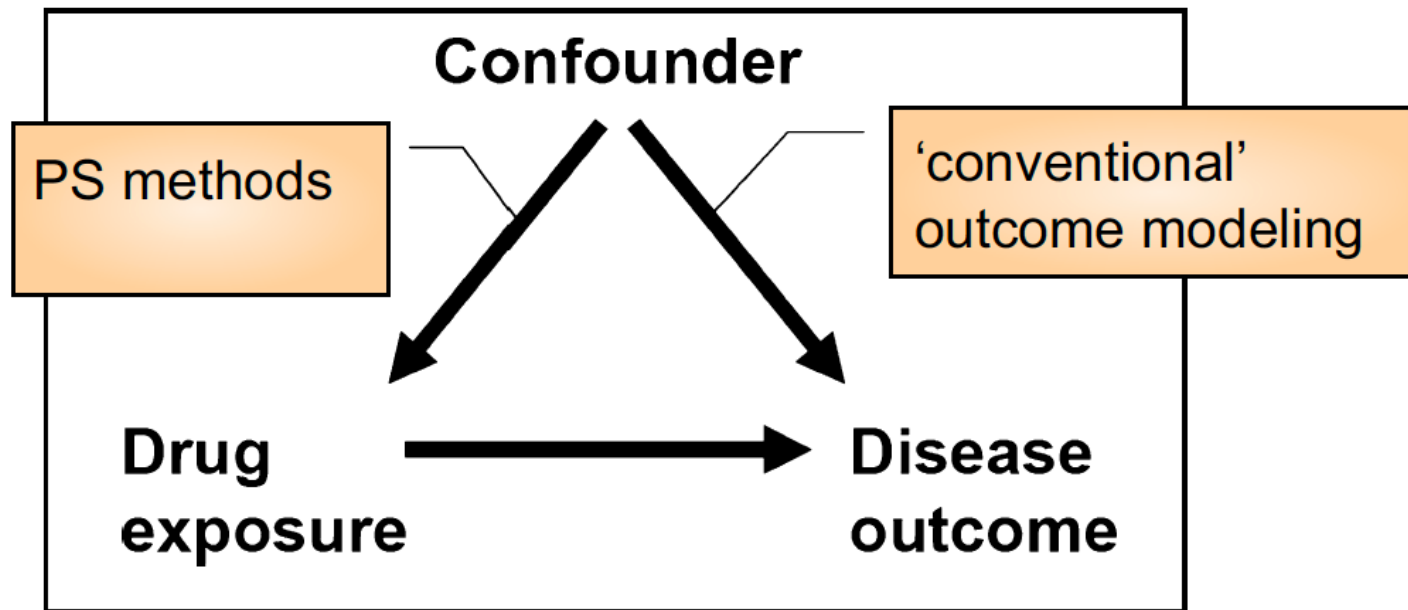


Fig. 1 Causal diagram for confounding. Both arrows need to be present to cause confounding. Removing one of the arrows removes confounding. This leads to essentially two general ways to control for confounding: we can remove the effect of the confounder on treatment (propensity score methods), or we can condition on the effect of the confounder on the outcome (outcome modelling). Both are dependent on having a good measure of the confounder available to the researcher.

Stürmer et al. J Intern Med. 2014;275:570-80

CONFOUNDING & INTERACTION

Confounding

MI	Coffee	
	No	Yes
Yes	10	40
No	490	460
	2.0%	8.0%

d=6%

Non-Smokers

MI	Coffee	
	No	Yes
Yes	8	7
No	472	413
	1.7%	1.7%

d=0%

Smokers

MI	Coffee	
	No	Yes
Yes	2	7
No	18	73
	10.0%	8.8%

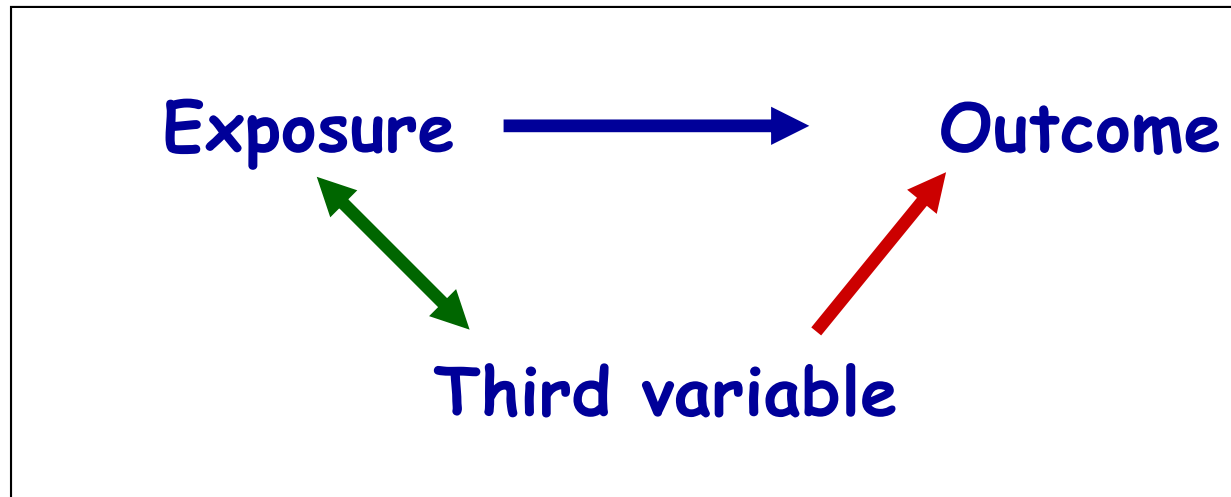
d=0%

Confounding

- A situation in which a measure of the effect of an exposure on risk is distorted because of the association of exposure with other factor(s) that influence the outcome under study.
- Criteria for confounding
 - Factor is associated with exposure
 - Factor is associated with disease in the absence of exposure
 - Factor is not in the causal path between exposure and outcome

Confounding

To be a confounding factor, two conditions must be met:



Be associated with exposure

- without being the consequence of exposure

Be associated with outcome

- independently of exposure (not an intermediary)

Interacction

MI	Smoking	
	No	Yes
Yes	9	34
No	491	466
	1.8%	6.8%

d=5%

Age < 45 years

MI	Smoking	
	No	Yes
Yes	4	6
No	296	294
	1.3%	2.0%

d=0.7%

Age >= 45 Years

MI	Smoking	
	No	Yes
Yes	5	28
No	195	172
	2.5%	14.0%

d=11.5%



Interaction & Subgroups

ISIS-2: Vascular death by Star signs

Geminis/Libra

	Aspirin	Placebo
Vascular Death	150	147
Total	1357	1442
	11.1%	10.2%

$p=0.42045$

$d=-0.9$

Other Star Signs

	Aspirin	Placebo
Vascular Death	654	868
Total	7228	7157
	9.0%	12.1%

$p<0.0001$

$d=3.1$

Interaction $p = 0.019$

Lancet 1988; 2: 349-60.

Changes from ISIS-2 results



***Lancet* 2005; 365: 1657-61**

Ferran.Torres@uab.cat

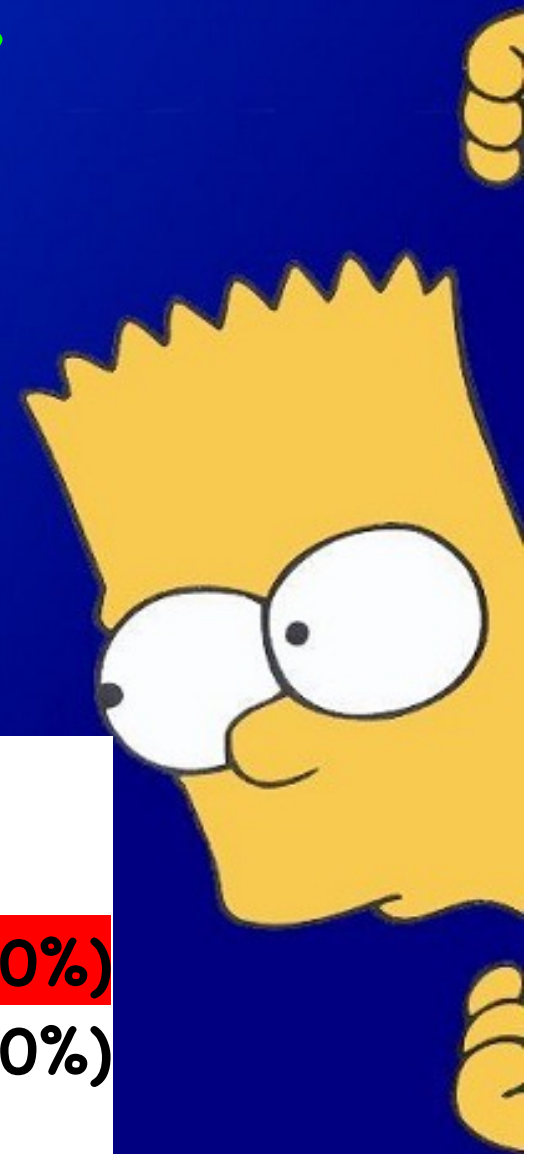
Simple Interaction Tests

- Tests for interaction of limited value when investigating subgroup differences
 - Low power to detect heterogeneity
 - Still have 5% or 10% false positive rate
 - Hypothesis testing not appropriate
- Estimates of size of interaction can be helpful to show what differences a trial can reliably estimate

Simpson's Paradox

Simpson EH. The interpretation of interaction in contingency tables. J R Stat Soc Series B Stat Methodol 1951; 13: 238–241.

		Experimental n (%)		Control n (%)	
ALL	Success	70	(70%)	60	(60%)
	Failure	30	(30%)	40	(40%)
		100		100	



Simpson's Paradox cont.

MALE

	Experimental n (%)	Control n (%)
Success	10 (33%)	24 (40%)
Failure	20 (67%)	36 (60%)
	30	60



FEMALE

	Experimental n (%)	Control n (%)
Success	60 (86%)	36 (90%)
Failure	10 (14%)	4 (10%)
	70	40

ALL

	Experimental n (%)	Control n (%)
Success	70 (70%)	60 (60%)
Failure	30 (30%)	40 (40%)
	100	100



Subgroups

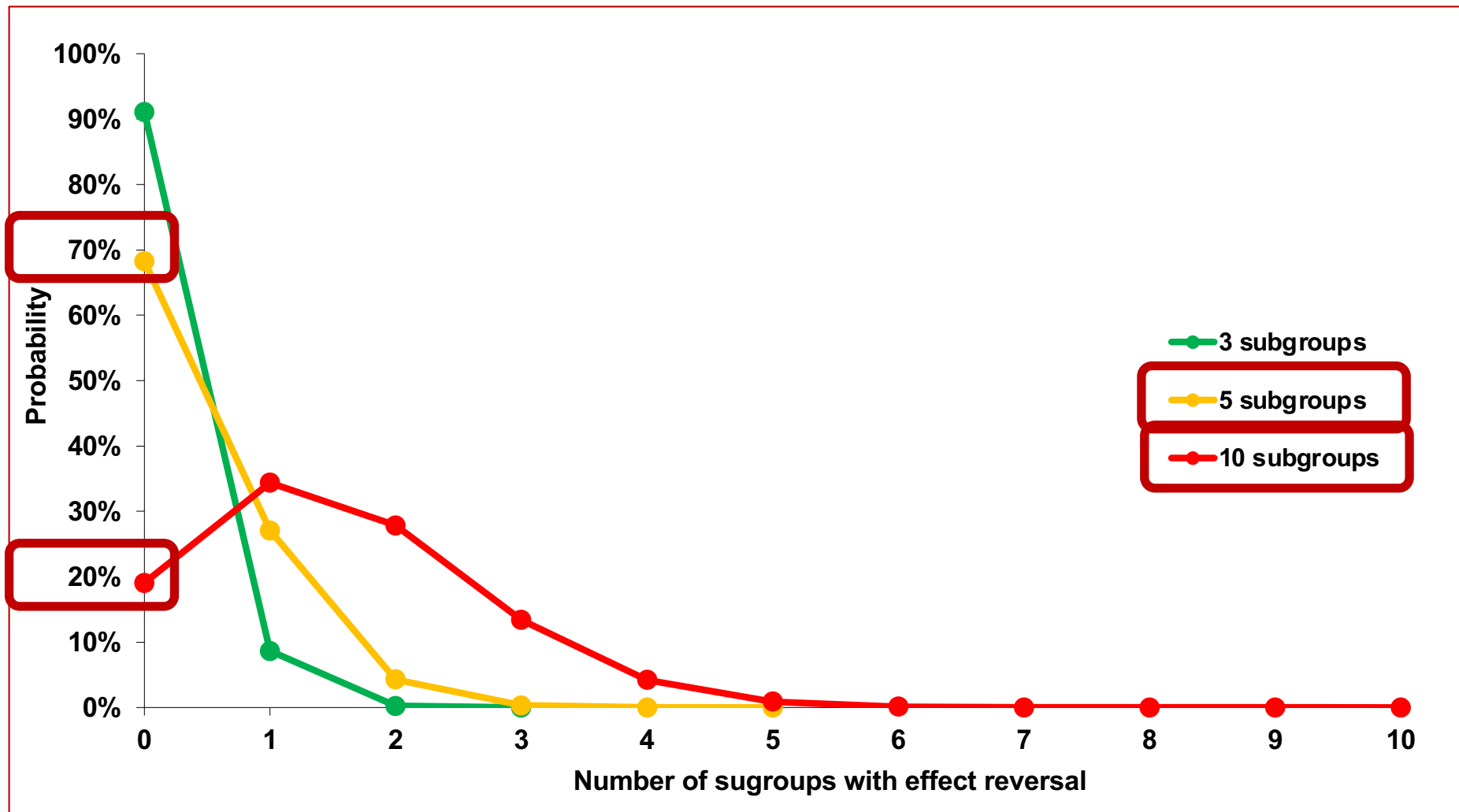
- Recommendations:
 - 1) Examine the global effect
 - 2) Test for the interaction
 - 3) Plan α adjustments for confirmatory analyses
 - 4) Some points which increase the credibility:
 - Pre-specification
 - Biologic plausibility

Observation	Refutation
Aspirin is ineffective in secondary prevention of stroke in women ^{29,30}	31
Antihypertensive treatment for primary prevention is ineffective in women ^{32,33}	34
Antihypertensive treatment is ineffective or harmful in elderly people ³⁵	36
Angiotensin-converting enzyme inhibitors do not reduce mortality and hospital admission in patients with heart failure who are also taking aspirin ³⁷	38
β blockers are ineffective after acute myocardial infarction in elderly people, ³⁹ and in patients with inferior myocardial infarction ⁴¹	40
Thrombolysis is ineffective >6 hours after acute myocardial infarction ⁴²	43
Thrombolysis for acute myocardial infarction is ineffective or harmful in patients with a previous myocardial infarction ⁴²	44
Tamoxifen citrate is ineffective in women with breast cancer aged <50 years ⁴⁵	46
Benefit from carotid endarterectomy for symptomatic stenosis is reduced in patients taking only low-dose aspirin due to an increased operative risk ⁴⁷	48
Amlodipine reduces mortality in patients with chronic heart failure due to non-ischaemic cardiomyopathy but not in patients with ischaemic cardiomyopathy ⁴⁹	50

Table 1: Examples of subgroup analyses that have shown apparently clinically important heterogeneity of treatment effect which has subsequently been shown to be false

***Lancet* 2005; 365: 176-86**

“Effect Reversals”, where the treatment effect is positive overall but numerically negative in some subgroups are to be expected in Phase III subgroup analyses



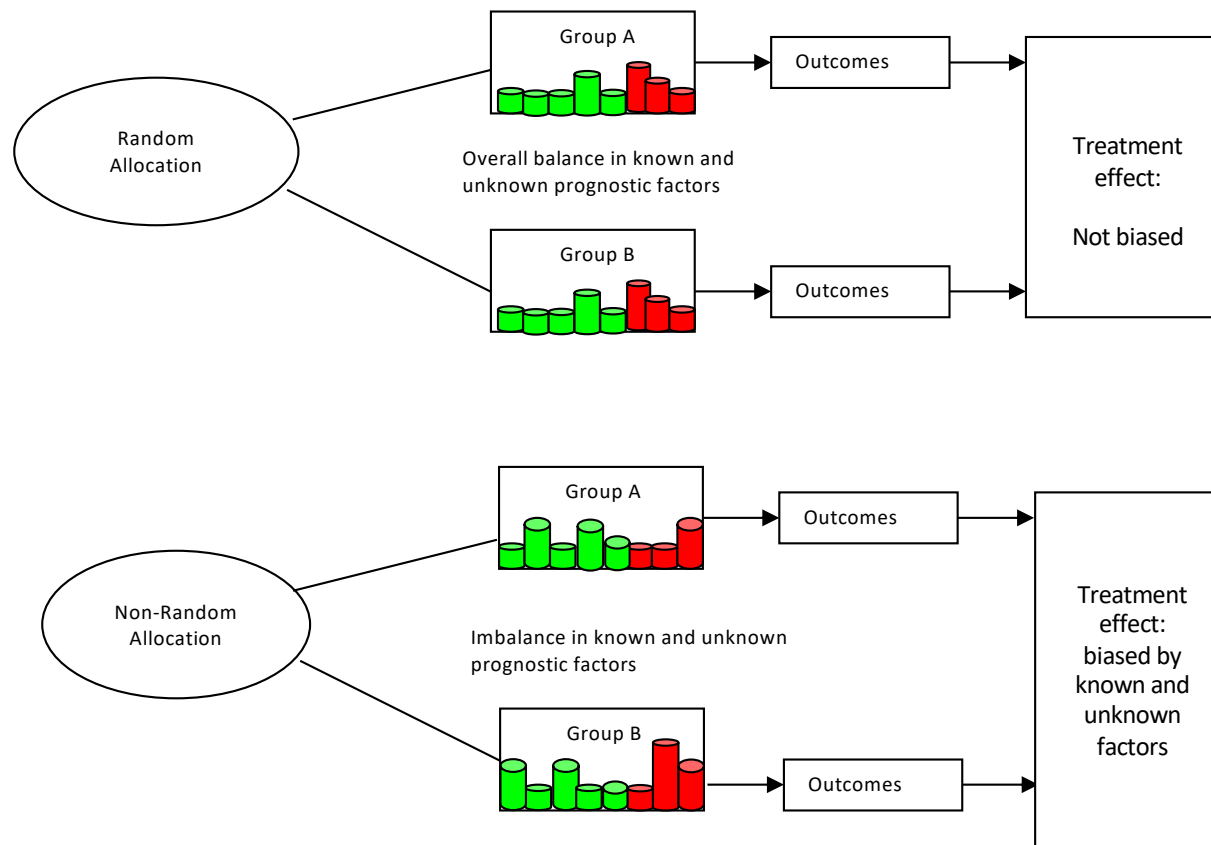
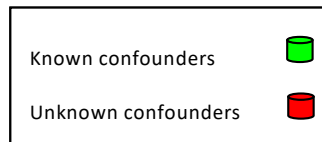


Fig. 1 Differential role of known and unknown confounders in observational studies and in randomized clinical trials. The distribution of unknown confounders is undetermined, but while they are on average balanced in randomized clinical trials, they are not in observational studies.

The 10% Rule for Confounding

The 10% Rule for Confounding

Most epidemiologists use a 10% difference as a "rule of thumb" for identifying the presence of confounding. The magnitude of confounding is the percent difference between the crude and adjusted measures of association, calculated as follows (for either a risk ratio or an odds ratio):

$$\text{Magnitude of Confounding (\%)} = \frac{RR_{crude} - RR_{adjusted}}{RR_{adjusted}} \times 100\%$$

If the % difference is 10% or greater, we conclude that there was confounding. If it is <10%, we conclude that there was little, if any, confounding.

For mortality rate ratios in Florida and Alaska:

$$\text{Magnitude of Confounding (\%)} = \frac{2.68 - 1.06}{1.06} = 152\%$$

Since 152% is much greater than 10%, the comparison of mortality rates was clearly confounded by differences in age distribution

Finally, note that the 10% rule of thumb for confounding is not rigid. Epidemiologists sometimes adjust for factors when the percent difference is less than 10%.

Method Favored by Biostatisticians

$$\text{Magnitude of confounding} = \frac{RR_{crude} - RR_{adjusted}}{RR_{crude}}$$

https://sphweb.bumc.bu.edu/otlt/MPH-Modules/PH717-QuantCore/PH717-Module11-Confounding-EMM/PH717-Module11-Confounding-EMM5.html#headingtaglink_5

Intercambiados!!

Distribution of Y	Link function name	Link function	Model	R
Normal	Identity	$g(\mu) = \mu$	$\mu = \mathbf{X}\beta$	<code>gaussian(link="identity")</code>
Binomial	Logit	$g(\mu) = \log\left(\frac{\mu}{1-\mu}\right)$	$\log\left(\frac{\mu}{1-\mu}\right) = \mathbf{X}\beta$	<code>binomial(link="logit")</code>
Poisson	Log	$g(\mu) = \log(\mu)$	$-\mu^{-1} = \mathbf{X}\beta$	<code>poisson(link="log")</code>
Exponential	Negative Inverse	$g(\mu) = -\mu^{-1}$	$\log(\mu) = \mathbf{X}\beta$	<code>Gamma(link="inverse")</code>

Variable distribution	family	link	Estimate of interest	regression type
binomial	binomial	logit	Odds Ratio: $\exp(\beta_x)$	Logistic
		log	Risk Ratio: $\exp(\beta_x)$	Log-Binomial
		Identity	Difference of %: β_x	Binomial
count data or rate data	poisson	Log	Risk/Rate Ratio: $\exp(\beta_x)$	Poisson
	negative.binomial	Log	Risk/Rate Ratio: $\exp(\beta_x)$	Negative Binomial

Table 2.1 Characteristics of some common univariate distributions in the exponential family[†]

	<i>Normal</i>	<i>Poisson</i>	<i>Binomial</i>	<i>Gamma</i>	<i>Inverse Gaussian</i>
<i>Notation</i>	$N(\mu, \sigma^2)$	$P(\mu)$	$B(m, \pi)/m$	$G(\mu, \nu)$	$IG(\mu, \sigma^2)$
<i>Range of y</i>	$(-\infty, \infty)$	$0(1)\infty$	$\frac{0(1)m}{m}$	$(0, \infty)$	$(0, \infty)$
<i>Dispersion parameter: ϕ</i>	$\phi = \sigma^2$	1	$1/m$	$\phi = \nu^{-1}$	$\phi = \sigma^2$
<i>Cumulant function: $b(\theta)$</i>	$\theta^2/2$	$\exp(\theta)$	$\log(1 + e^\theta)$	$-\log(-\theta)$	$-(-2\theta)^{1/2}$
<i>$c(y; \phi)$</i>	$-\frac{1}{2} \left(\frac{y^2}{\phi} + \log(2\pi\phi) \right)$	$-\log y!$	$\log \binom{m}{my}$	$\nu \log(\nu y) - \log y - \log \Gamma(\nu)$	$-\frac{1}{2} \left\{ \log(2\pi\phi y^3) + \frac{1}{\phi y} \right\}$
<i>$\mu(\theta) = E(Y; \theta)$</i>	θ	$\exp(\theta)$	$e^\theta/(1 + e^\theta)$	$-1/\theta$	$(-2\theta)^{-1/2}$
<i>Canonical link: $\theta(\mu)$</i>	identity	log	logit	reciprocal	$1/\mu^2$
<i>Variance function: $V(\mu)$</i>	1	μ	$\mu(1 - \mu)$	μ^2	μ^3

[†]The mean-value parameter is denoted by μ , or by π for the binomial distribution.

The parameterization of the gamma distribution is such that its variance is μ^2/ν .

The canonical parameter, denoted by θ , is defined by (2.4). The relationship between μ and θ is given in lines 6 and 7 of the Table.

Monographs
on Statistics and
Applied Probability 37

Generalized Linear Models

SECOND EDITION

P. McCullagh and
J.A. Nelder FRS

Modelling: What is the Structure of the Relationship?

Is a linear model appropriate? Are there alternatives?

- Other nonlinearities
 - Quadratic
 - Exponential
 - Logarithmic – post-course
 - Etc..
- Qualitative variables
- Trend test
- Interactions

		Grouping variable		Univariable logistic regression		
		no n=93	yes n=8	Effect	p-value	OR [95%CI]
edad	(years)	median [25th-75th percentile]	median [25th-75th percentile]	Lineal per 10 years	0.0456	exemple 1.223 [1.111 - 2.222]
				Standardized lineal per 1 SD	0.0456	exemple 2.4 [1.111 - 2.222]
				Quadratic per 10 years		
				Tertiles Trend test		
				1st		1 (reference)
				2nd		...
sexo	Male	n %	n %		0.0345	1 (reference)
	Female	n %	n %			2.555 [1.222 - 4.666]
Var 1	...	(segons tipus variable, el mateix que edad o sexo)	(segons tipus variable, el mateix que edad o sexo)		...	...
Var 2	...	(segons tipus variable, el mateix que edad o sexo)	(segons tipus variable, el mateix que edad o sexo)		...	...

HR per 1 unit change (from model)									
	Parameter Estimate	Standard Error		HR	LL	UL			
MELD_Na	0.1315	0.0141		1.141	1.109	1.173			
LFABP_Urine	0.0026	0.0007		1.003	1.001	1.004			

HR per 1 unit change (from model)									
	Parameter	Standar		HR	LL	UL			
	Estimate	d Error							
MELD_Na	0.1315	0.0141		1.141	1.109	1.173			
LFABP_Urine	0.0026	0.0007		1.003	1.001	1.004			
HR per 10 unit change (transforming data dviding by the unit-factor)									
Effect	Parameter	Standar		HR	LL	UL			
	Estimate	d Error		UNITS	UNITS	UNITS			
MELD_Na_10	1.3154	0.1414		3.726	2.824	4.916			
LFABP_Urine_10	0.0261	0.0075		1.026	1.011	1.042			

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Units using Betas				EXP(10*Beta)					
Effect	Parameter Estimate	Standard Error	UNITS	HR UNITS	LL UNITS	UL UNITS			
MELD_Na	0.1315	0.0141	10	3.726	2.824	4.916			
LFABP_Urine	0.0026	0.0007	100	1.298	1.121	1.502			

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LFABP_Urine	0.0026	0.0007	100	1.298	1.121	1.502			
Units using HR Potencia(HR;10)									
				HR	LL	UL	UNITS	HR UNITS	LL UNITS
								UL UNITS	
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Units using HR				Potencia(HR;10)						
				HR	LL	UL	UNITS	HR UNITS	LL UNITS	UL UNITS
MELD_Na				1.141	1.109	1.173	10	3.726	2.824	4.916
LFABP_Urine				1.003	1.001	1.004	100	1.298	1.121	1.502

Recommended: use the SD as the factor for comparing continuous variables

TRIPOD Statement

Moons KG, Altman DG, Reitsma JB, Ioannidis JP, Macaskill P, Steyerberg EW, Vickers AJ, Ransohoff DF, Collins GS. Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration. *Ann Intern Med*. 2015 Jan 6;162(1):W1-73. doi: 10.7326/M14-0698. PMID: 25560730.

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Youden WJ. Index for rating diagnostic tests. Cancer. 1950 Jan;3(1):32-5. doi: 10.1002/1097-0142(1950)3:1<32::aid-cnrcr2820030106>3.0.co;2-3. PMID: 15405679.

Calibration

Austin PC, Steyerberg EW. Graphical assessment of internal and external calibration of logistic regression models by using loess smoothers. Stat Med 2014;33:517–535.

Nattino G, Finazzi S, Bertolini G. A new calibration test and a reappraisal of the calibration belt for the assessment of prediction models based on dichotomous outcomes. Stat Med. 2014 Jun 30;33(14):2390-407. doi: 10.1002/sim.6100. Epub 2014 Feb 4.

Net reclassification index (NRI)

Pencina MJ, D'Agostino RB Sr, D'Agostino RB Jr, Vasan RS. Evaluating the added predictive ability of a new marker: from area under the ROC curve to reclassification and beyond. *Stat Med*. 2008 Jan 30;27(2):157-72; discussion 207-12. doi: 10.1002/sim.2929. PMID: 17569110.

Resampling techniques: bootstrapping, jack-knife, or cross-validation

En Steyerberg's book

Decision curve analysis (DCA)

Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making*. 2006 Nov-Dec;26(6):565-74. doi: 10.1177/0272989X06295361. PMID: 17099194; PMCID: PMC2577036.

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